MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6112

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document andt the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, subminiature, high Mu

Outline --- 3-1 (EIA)

Base --- E8-10

Envelope --- T3

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2a	2g	h	2k	1k	h	1g	1a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk/k	Rg/g	Ib/a	Pp/a	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	W	°C	ft
Maximum:	6.6	165	0, -55	200	---	1.1	3.3	0.30	+220	Note 1
Minimum:	6.0	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	0	1,500	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	Limits MIN	Limits MAX	UNIT
	Quality conformance inspection, part 1							
1256	Electrode current (1) (anode)	See notes 3 and 4	0.4	II	Ib	0.5	1.10	mAdc
1256	Electrode current (2) (anode)	Ec = -2.8 Vdc; Rk = 0 (see note 3)	0.4	II	Ib	---	50	µAdc
1266	Total grid current	Eb = 150 Vdc; Ec = 0; Rk/k = 820 ohms; Rg/g = 1.0 Meg (see notes 3 and 4)	0.4	II	Ic	0	-0.3	µAdc
1301	Heater current		0.4	II	If	280	320	mA
1306	Transconductance (1)	See note 3	0.4	II	Sm	1,500	2,100	µmhos
1336	Heater-cathode leakage	See note 3	0.4	II	Ihk	---	5.0	µAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	See note 3	2.5	I	---	---	---	---
1031	Low-frequency vibration	Rp = 10,000 ohms; F = 40 Hz; 15 G (see note 3)	2.5	I	Ep	---	25	mVac
1246	Audio frequency noise	Esig = 45 mVac; Rg = 0.5 Meg; Rp = 0.2 Meg; Rk = 750 ohms; (see note 5)	2.5	I	EB	---	17	vu
1266	Grid emission	Ef = 7.5 V; Ec = -4.0 Vdc; Eb = 150 Vdc; Rk = 0; Rg/g = 1.0 Meg (see notes 3 and 6)	2.5	I	Ic	0	-0.5	µAdc
1306	Transconductance (2)	Ef = 5.7 V (see note 3)	2.5	I	ΔSm Ef	---	15	%
1316	Amplification factor	See note 3	---	---	Mu	60	80	---
1321	AC amplification	Ebb = 100 Vdc; Ecc = 0; Esig = 0.2 Vac; Rk = 0 (see note 3)	---	---	Ep	8.0	---	Vac
1331	Direct-interelectrode capacitance	No shield (see note 3) No shield (see note 3) No shield; section 1 No shield; section 2 No shield No shield	6.5	Code F	Cgp Cin Cout Cout Cgg Cpp	0.8 1.30 0.16 0.21 --- ---	1.20 2.10 0.30 0.35 0.014 0.80	pF pF pF pF pF pF
1116	Lead fatigue		2.5	Code F	---	---	---	---
2126	Envelope strain		4.0	I	---	---	---	---

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 2 -Continued							
1041	Shock	450 G; Ehk = +100 Vdc; Rg = 0.1 Meg (see note 7)	---	---	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 min, 60 max	6.5	See note 8	---	---	---	---
---	Post-shock and vibration-fatigue test end points:							
1031	Low-frequency vibration		---	---	Ep	---	100	mVac
1336	Heater-cathode leakage		---	---	Ihk	---	20	µAdc
1306	Change in transconductance (1) of individual tubes		---	---	ΔSm t	---	20	%
1105	Permanence of marking							

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1 ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 7.0 V; 1 min "on", 4 min "off"; Ehk = 140 Vac; Ec = Eb = 0	---	---	---	---	---	---	---	---
1516	Stability life	Eb = 150 Vdc; Ehk = +200 Vdc; Rg/g = 1.0 Meg; Rk/k = 820 ohms; TA = room (see note 3)	---	---	---	---	---	---	---	---
---	Stability life-test end point:									
1306	Change in transconductance (1) of individual tubes		---	---	---	---	ΔSm t	---	10	%
1501	Intermittent life (room temperature)	Stability life test, or equivalent conditions; TA = room (see notes 9 and 11)	---	---	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL PERCENT DEFECTIVE	MIL LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 -Continued									
---	Intermittent life-test end point (room temperature (500 hours)									
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg
---	Intermittent life-test end point (room temperature (1,000 hours)									
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
1501	Intermittent life (high temperature)	Stability life-test conditions; TE = +220° C (min) (see notes 3 and 10)	---	---	---	---	---	---	---	---
---	Intermittent life-test end points:(500 hours) (high temperature)									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current		---	---	1	3	Ic	0	-0.9	µAdc
1301	Heater current		---	---	1	3	If	276	328	mA
1306	Change in transconductance (1) of individual tubes		---	---	1	3	ΔSm t	---	20	%
1306	Transconductance (2)		---	---	1	3	ΔSm Ef	---	15	%
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	10	µAdc
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg
1306	Transconductance (1) average change		---	---	---	---	Avg ΔSm t	---	15	%
---	Total defectives		---	---	3	6	---	---	---	---
---	Intermittent life-test end points:(1,000 hours) (high temperature)									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current		---	---	1	3	Ic	0	-0.9	µAdc
1301	Heater current		---	---	1	3	If	276	328	mA
1306	Change in transconductance (1) of individual tubes		---	---	1	3	ΔSm t	---	25	%
1306	Transconductance (2)		---	---	1	3	ΔSm Ef	---	20	%
1336	Heater-cathode leakage		---	---	1	3	Ihk	---	10	µAdc
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
---	Total defectives		---	---	4	8	---	---	---	---

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Revision letter are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the change.

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NOTES: -Continued

3. Test each unit separately.

4. This test to be performed at the conclusion of the holding period.

5. Tie 1k to 2k; 1g to 2g; and 1a to 2a.

6. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions specified below. The 3-minute test is not permitted. Test as specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Ec1	Eb	Rk	Rg
V	Vdc	Vdc	Ohms	Meg
7.5	0	150	820	1.0

7. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron type short indicator is used.

8. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

10. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

11. Early release may be made at 500 hours providing no lot failure due to this life test or the 1,000-hour intermittent life test (high temperature) has occurred in the preceding three consecutive lots, and the lot under consideration is acceptable at 500 hours. One lot failing the 1,000-hour intermittent (high temperature) or the 1,000-hour intermittent (room temperature) life test shall result in loss of eligibility for the early release procedure. All intermittent (room temperature) life-test samples shall be operated to 1,000 hours.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 17, 99
DLA - ES

User activities:
Army -
Navy - AS, MC, CG, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA-ES

(Project 5960-3175)

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